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## DEVICES AND METHODS FOR TREATING TESTICULAR PAIN

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### Abstract

The present disclosure relates to a method of treating testicular pain. The method includes externally coupling an electrical energy source to a pelvic region of a subject, coupling a vibration mechanism to the pelvic region of the subject, delivering electrical energy to the pelvic region, and applying vibrations to the pelvic region.

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### Background/Summary

#### TECHNICAL FIELD

[0001] The present disclosure relates to devices and methods for treating testicular pain, and in particular, to devices and methods that deliver electrical energy and vibration to a pelvic region to treat testicular pain.

#### BACKGROUND

[0002] Testicular pain may be caused by a variety of injuries or disorders in men. For example, iliohypogastric neuralgia, ilioinguinal neuralgia, genitofemoral neuralgia, chronic groin pain, chronic testicular pain (CTP), post vasectomy pain, hypogastric nerve injury, anterior extrusions, and other pain originating from the testicles, groin, or abdomen are common reasons for referral to a urological specialist. The testicular pain experienced by a subject may be unilateral or bilateral, constant or intermittent, spontaneous or exacerbated by physical activities and pressure. Pain may remain localized in the scrotum or radiate to throughout the pelvic region to the groin or perineum, or extend to the back or legs.

#### SUMMARY

[0003] According to the present disclosure, devices and methods for non-invasively treating testicular pain using electrical and vibrational stimulation may be adapted to a wide variety of physiologies of subjects to treat pain.

[0004] In a first example aspect, a method of treating testicular pain may include coupling an electrical energy source to a pelvic region of the subject, coupling a vibration mechanism to the pelvic region of the subject, delivering electrical energy to the pelvic region, and applying vibrations to the pelvic region.

[0005] In a second example aspect, a method of treating testicular pain in a subject may include coupling a stimulation device to a pelvic region of the subject. The stimulation device may include an electrical contact configured for delivering electrical energy and a vibration mechanism configured for delivering vibrations. The method may include delivering, via the stimulation device, electrical energy to the subject. The method may include delivering, via the stimulation device, vibrations to the subject.

[0006] In a third example aspect, a wearable device for treating testicular pain may include a first stimulation contact configured for delivering an electrical pulse to a pelvic region of a subject, and a second stimulation contact configured for delivering vibrations to the pelvic region of the subject.

[0007] In accordance with any one of the first, second, and third example aspects, a method for treating testicular pain and a wearable stimulation device to treat testicular pain may include any one or more of the following forms.

[0008] In one form, coupling the electrical energy source may include externally engaging the electrical energy source with a scrotum of the subject.

[0009] In some forms, coupling the vibration mechanism may include engaging the vibration mechanism with a scrotum of the subject.

[0010] In other forms, coupling the electrical energy source and coupling the vibration mechanism may include connecting a stimulation device to the pelvic region of the subject.

[0011] In some forms, the stimulation device may include an electrode and the vibration mechanism.

[0012] In another form, coupling the stimulation device may include attaching a cuff at least partially around a scrotum of the subject.

[0013] In some forms, the cuff may include the electrode and the vibration mechanism.

[0014] In some forms, the electrode and the vibration mechanism may be arranged to engage the scrotum when the cuff is attached to the scrotum.

[0015] In yet another form, the method may include positioning the vibration mechanism in a first area of the pelvic region and positioning the electrode in a second area of the pelvic region.

[0016] In one example, coupling the electrical energy source may include adhering an electrode to the subject.

[0017] In some examples, delivering the electrical energy may include delivering an electrical pulse at a frequency in a range of approximately 0 Hz to approximately 150 Hz.

[0018] In other examples, applying vibrations may include applying vibrations having a frequency in a range of approximately 0 Hz to approximately 50 Hz.

[0019] In another example, applying vibrations and delivering the electrical energy may include

delivering an electrical pulse and applying vibrations to a testicular cord, a cremasteric muscle, an inguinal canal, a genitofemoral nerve, an ilioinguinal nerve, and/or an iliohypogastric nerve of the subject.

[0020] In yet another example, coupling the stimulation device may include engaging one or more of the electrical contacts and the vibration mechanism to a scrotum of the subject.

[0021] In some examples, coupling the stimulation device may include positioning a second electrical contact of the stimulation device in an area of the pelvic region that is different than the scrotum of the subject.

[0022] In one aspect, coupling the stimulation device may include fastening a cuff of the stimulation device around a scrotum of the subject.

[0023] In another aspect, the electrical contact and the vibration mechanism may be integrated with the cuff of the stimulation device.

[0024] In another aspect, the method may include adjusting, via a controller, an intensity of a deliverable electrical energy and/or deliverable vibrations.

[0025] In some aspects, the controller may be operatively coupled to the electrical contact and to the vibration mechanism.

[0026] In yet another aspect, adjusting may include adjusting a stimulation setting using a personal electronic device.

[0027] In some aspects, the controller may be communicatively coupled to the personal electronic device.

[0028] In other aspects, a flexible band may be arranged for removably attaching the first stimulation contact and the second stimulation contact to a scrotum of the subject.

[0029] In some aspects, the first stimulation contact and second stimulation contact may be coupled to the flexible band.

[0030] In one form, a controller may be operatively coupled to the first stimulation contact and the second stimulation contact.

[0031] In some forms, the controller may include a processor, a memory communicatively coupled to the processor and storing executable instructions that, when executed by the processor, causes the processor to receive data transmitted by a remote communication device, send a signal to a generator coupled to the first stimulation contact, adjust an energy parameter of the electrical pulse generated by the generator, send a signal to a vibratory element of the second stimulation contact, and adjust an intensity parameter of vibrations generated by the vibratory element.

[0032] In one form, externally coupling an electrical energy source may include engaging the electrical energy source with a scrotum of the subject.

[0033] In some forms, externally coupling the electrical energy source may include coupling an electrical stimulation device to a scrotum of the subject.

[0034] In other forms, wherein the electrical energy source may include a first electrode and a second electrode of the electrical stimulation device.

[0035] In another form, coupling the electrical stimulation device may include attaching a cuff at least partially around the scrotum of the subject.

[0036] In some forms, the cuff may include an electrode positioned to engage the scrotum when the cuff is attached to the scrotum.

[0037] In yet another form, externally coupling the electrical energy source may include coupling an electrode of an electrical stimulation device to a scrotum of the subject.

[0038] In one example, externally coupling the electrical energy source may include coupling a second electrode to a different area of the pelvic region.

[0039] In some examples, externally coupling the electrical energy source may include adhering an electrode to a pelvic region adjacent to an inguinal canal and/or cremasteric muscle.

[0040] In other examples, delivering the electrical energy may include delivering an electrical pulse to the inguinal canal and/or cremasteric muscle of the subject.

[0041] In yet another example, delivering the electrical energy may include delivering an electrical pulse to a testicular cord, a cremasteric muscle, an inguinal canal, a genitofemoral nerve, an ilioinguinal nerve, and/or an iliohypogastric nerve of the subject.

[0042] In one aspect, the method may include adjusting a frequency of the electrical pulse being delivered.

[0043] In some aspects, the frequency may be in a range of approximately 0 Hz to approximately 150 Hz.

[0044] In some aspects, the method may include adjusting a pulse width of the electrical pulse being delivered.

[0045] In one aspect, the pulse width may be in a range of approximately 50 microseconds to approximately 200 microseconds.

[0046] In other aspects, externally coupling may include coupling a first electrode of an electrical stimulation device to a first area of the pelvic region and coupling a second electrode of the electrical stimulation device to a second area of the pelvic region.

[0047] In another aspect, coupling the stimulation device may include positioning a second contact in proximity to an inguinal canal of the subject.

[0048] In yet another aspect, coupling the stimulation device may include positioning a second contact of the stimulation device in an area different than the scrotum in a pelvic region of the subject.

[0049] In one form, the method may include positioning a cuff of the stimulation device around the scrotum of the subject.

[0050] In other forms, the contact may be integrated with the cuff of the stimulation device.

[0051] In some forms, delivering electrical energy may include delivering an electrical pulse to the subject.

[0052] In other forms, the method may include adjusting a setting of the electrical pulse.

[0053] In another form, adjusting the setting may include communicating with a controller using a remote communication device.

[0054] In some forms, the controller may be communicatively coupled to the electrical energy generator.

[0055] In yet another form, the method may include adjusting a frequency of the one or more electrical pulses being delivered.

[0056] In one example, the frequency may be in a range of approximately 0 Hz to approximately 150 Hz.

[0057] In one example, the method may include adjusting a pulse width of the electrical pulse being delivered.

[0058] In some examples, the pulse width may be in a range of approximately 50 microseconds to 200 microseconds.

[0059] In some examples, delivering the electrical energy may include delivering an electrical pulse to an inguinal canal of the subject.

[0060] In other examples, delivering the electrical energy may include delivering an electrical pulse to a genitofemoral nerve of the subject.

[0061] In yet another example, delivering the electrical energy may include delivering an electrical pulse to an ilioinguinal nerve of the subject.

[0062] In one aspect, delivering the electrical energy may include delivering an electrical pulse to an iliohypogastric nerve of the subject.

[0063] In some aspects, delivering the electrical energy may include delivering an electrical pulse to a cremasteric muscle of a testicular cord.

[0064] In other aspects, the device may include a garment defining a pocket sized to receive the generator.

[0065] In some aspects, the generator may be insertable in the pocket of the garment.

[0066] In another aspect, the stimulation contact may include an electrode configured for delivering an electrical pulse.

[0067] In yet another aspect, the device may include a flexible band arranged for removably attaching to a scrotum of a subject.

[0068] In one form, the electrical stimulation source may be coupled to the band.

[0069] In one form, the band may be arranged to form a circular cuff sized to fit around a scrotum of a subject.

[0070] In some forms, a controller may be operatively coupled to the generator.

[0071] In another form, the controller may include a processor and a memory communicatively coupled to the processor and storing executable instructions that, when executed by the processor, causes the processor to receive data transmitted by a remote communication device, and send a signal to the generator.

[0072] In other forms, the instructions may cause the processor to adjust an energy parameter of the energy generated by the generator.

[0073] In one form, applying vibrations may include applying vibrations to a testicular cord of the human subject.

[0074] In another form, applying vibrations may include applying vibrations to a cremasteric muscle of the human subject.

[0075] In some forms, applying vibrations may include applying vibrations to an inguinal canal of the human subject.

[0076] In other forms, applying vibrations may include applying vibrations to an ilioinguinal nerve of the human subject.

[0077] In yet another form, applying vibrations may include applying vibrations to an iliohypogastric nerve of the human subject.

[0078] In one example, applying vibrations may include applying vibrations to one or more of a genitofemoral nerve and a pudendal nerve of the human subject.

[0079] In some examples, coupling the vibration mechanism may include engaging the vibration mechanism with a scrotum of the human subject.

[0080] In another example, the method may include coupling a second vibration mechanism to another area of the pelvic region of the human subject.

[0081] In other examples, the method may include adjusting a frequency of the vibrations being applied, wherein the frequency is in a range of approximately 15 Hz to approximately 55 Hz.

[0082] In yet another example, applying the vibration device may include positioning a second vibration mechanism of the vibration device in a second target location of the pelvic region of the human subject.

[0083] In one aspect, applying the vibration device may include fastening a strap around the human subject.

[0084] In another aspect, the vibration mechanism may be supported by the strap.

[0085] In another aspect, applying the vibration device may include dressing the human subject with an undergarment.

[0086] In some aspects, the undergarment may be configured to support to the vibration mechanism.

[0087] In yet another aspect, the method may include inserting the vibration mechanism inside a pocket of the undergarment.

[0088] In some aspects, the method may include communicating with a controller using a remote communication device to adjust a frequency of a vibratory element of the vibration mechanism, wherein the controller is communicatively coupled to the vibratory element.

[0089] In other aspects, the vibration mechanism may include a second vibratory element.

[0090] In some forms, the controller may be configured to separately operate the vibratory element and the second vibratory element at different frequencies.

[0091] In one form, the garment may include a flexible band arranged for removably attaching to a pelvic region of the human subject.

[0092] In another form, the vibration mechanism may be coupled to the flexible band.

[0093] In other forms, a controller may be operatively coupled to the vibratory element.

[0094] In some forms, the controller may include a processor, a memory communicatively coupled to the processor and storing executable instructions that, when executed by the processor, causes the processor to receive data transmitted by a remote communication device, send a signal to the vibratory element, and adjust an intensity parameter of a vibration applied by the vibratory element.

#### Definitions

[0095] As used herein, the terms “top,” “bottom,” “upper,” “lower,” “above,” and “below” are used to provide a relative relationship between structures. The use of these terms does not indicate or require that a particular structure must be located at a particular location in the apparatus.

[0096] Some examples may be described using the expression “coupled” and “connected” along with their derivatives. For example, some arrangements may be described using the term “coupled” to indicate that two or more elements are in direct physical or electrical contact. The term “coupled,” however, may also mean that two or more elements are not in direct contact with each other, but yet still co-operate or interact with each other. The examples described herein are not limited in this context.

[0097] As used herein, the term “pelvic region” or “pelvic/scrotal region” may be used to refer to right, left, or both sides of one or more of a lower pelvic region, scrotum, perineum, epididymis, vas deferens, testicle, testicular cord (or spermatic cord), cremasteric muscle(s), inguinal ring, inguinal canal, genitofemoral nerve, ilioinguinal nerve, iliohypogastric nerve, and pudendal nerve.

[0098] Examples of the present disclosure can include one or more of the following advantages.

[0099] The disclosed methods and devices may advantageously modulate the electrical or vibrational signals coming into the targeted areas (e.g., the testicular cord, including the testicular cremasteric muscles, inguinal canal, ilioinguinal nerve, the iliohypogastric nerve, and/or iliohypogastric nerve) and downregulate the sensitivity of the spinal cord/brain of the subject, such that the normal pain signals coming in will no longer be sufficient to register as pain.

[0100] In some examples, the disclosed methods and devices may be used to treat scrotal pain, chronic pelvic pain, meralgia paraesthetica, and cramping discomfort of irritable bowel syndrome.

[0101] In some examples, the disclosed methods and devices may provide alternatives to pharmacologic and surgical therapies.

[0102] In some examples, the disclosed devices may be integrated with a wearable garment or accessory to provide a discreet device that is low profile and quiet.

[0103] In some examples, the disclosed methods and devices may exhaust the ability of the targeted nerves to fire, which can occur with repeated stimulation.

[0104] In some examples, the disclosed methods and stimulation devices may be adapted to treat a patient population having subjects in a wide range of weight classes and with scrotal lengths and sizes.

[0105] In some examples, the wearable stimulation device of the present disclosure has an integrated, adjustable coupler that permits attaching to a scrotum according to the subject's scrotal physiology or comfort. The adjustable coupler can be tightened or loosened easily.

[0106] In another example, the stimulation device with a flexible electrode contact pad may be placed under a subject's panniculus or other areas of the body to effectively treat testicular pain. The electrodes are flexible and come in a variety of shapes and sizes to best suit a subject's physiology and comfort.

[0107] Other features and advantages of the present disclosure will be apparent from the following detailed description, figures, and claims.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0108] FIG. 1 is a front view of an example stimulation device for treating testicular pain using electrical energy and vibration in accordance with the teachings of the present disclosure;

[0109] FIG. 2 is a front view of the stimulation device of FIG. 1 coupled to a pelvic/scrotal region of a human subject;

[0110] FIG. 3A is a front view of an example stimulation assembly for treating testicular pain in accordance with the teachings of the present disclosure;

[0111] FIG. 3B is a cross-sectional side view of a schematic drawing of a vibration mechanism of the assembly of FIG. 3A;

[0112] FIG. 4 is a back view of another example stimulation device for treating testicular pain in accordance with the teachings of the present disclosure;

[0113] FIG. 5 is a front view of the stimulation device of FIG. 4;

[0114] FIG. 6 is a front view of the stimulation device of FIG. 4 shaped to form a circular cuff;

[0115] FIG. 7 is a front view of the stimulation device of FIG. 4 coupled to a pelvic/scrotal region of a human subject;

[0116] FIG. 8 is a front view of a wearable system for treating testicular pain in accordance with the teachings of the present disclosure;

[0117] FIG. 9 is a front view of another wearable assembly for treating testicular pain in accordance with the teachings of the present disclosure;

[0118] FIG. 10 is a front view of another wearable assembly for treating testicular pain in accordance with the teachings of the present disclosure;

[0119] FIG. 11 is a front view of another wearable assembly for treating testicular pain in accordance with the teachings of the present disclosure;

[0120] FIG. 12 is a front view of another wearable assembly for treating testicular pain in accordance with the teachings of the present disclosure; and

[0121] FIG. 13 is a schematic drawing of a control system that can be used to perform operations in accordance with the teachings of the present disclosure.

### DETAILED DESCRIPTION

[0122] According to the present disclosure, devices and methods for non-invasively treating testicular pain may be adapted to a wide variety of physiologies of subjects to treat pain in the pelvic region, and specifically, near, above, and/or on the scrotum.

[0123] In FIG. 1, a first example stimulation device 10 is configured for external stimulation of a pelvic/scrotal region of a subject. The stimulation device 10 includes an electrical signal generator 14, a plurality of electrical contacts 18, 26, a battery 30 coupled to the generator 14, and a vibration mechanism 28. The generator 14, battery 30, and a controller 31 are disposed in a housing 32 of the device 10.

[0124] The first and second electrical contacts 18, 26 are electrically coupled to the generator 14 and the power source (e.g., a battery) 30, and the vibration mechanism 28 is coupled to the battery 30. In this example, the first and second electrical contacts 18, 26 are electrodes. The controller 31 is operatively coupled to the stimulation contacts 18, 26, 28 (e.g., both electrical and vibrational contacts) through a wired connection. Each of the electrical contacts 18, 26 includes a wire 34 having a plug 38 at a first end and a contact pad 42, which delivers the electrical energy to the subject, at an opposing end. Each wire 34 of the electrical contacts 18, 26 electrically couples the electrical contact 18, 26 to the generator 14. The vibration mechanism 28 also includes a wire and a plug 38 configured for connecting to the housing 32. In some examples, the vibration mechanism 28 may also be coupled to the generator 14.

[0125] The electrical contacts 18, 26 and vibration mechanism 28 are arranged for placement at

various areas on or near a pelvic region **46** of a subject **48** to deliver stimulating energy, in the form, for example, of electrical pulses and vibrations, to the subject **48** to treat testicular pain. The electrical contacts **18**, **26** and the vibration mechanism **28** may be used to apply electrical energy and vibration anywhere from an inguinal ring to a scrotum of a subject.

[0126] The vibration mechanism **28** includes a housing **40**, a vibratory element **44** disposed in the housing **40**, the wire **34**, and the plug **38**. The vibration mechanism **28** is configured to apply a massage-like input to a target nerve, muscle, or tissue. The vibratory element **44** is a DC motor that generates vibrations, such as an Eccentric Rotating Mass (“ERM”) vibration motor, which uses an unbalanced load to generate vibrations. However, in other examples, the vibratory element **44** may be a pancake or shaftless motor, a piezoelectric vibration generator, or other suitable means for producing vibrations. The vibratory element **44** is centrally located relative to the vibration mechanism housing **40**. However, in this or other examples, the vibratory element **44** may be placed at any location within the housing **40**.

[0127] The vibration mechanism housing **40** is a thin, pliable membrane capable of protecting the internal components from sweat or other moisture, while also delivering vibrations to the subject **48**. One or more layers of material (e.g., material **199** disposed in vibration mechanism housing **140** of FIG. 3B) may be disposed in the housing **40** that contribute (e.g., amplify, dampen, etc.) to a particular vibration massage or treatment. The housing **40** may be composed of and/or contain materials that are both capable of transmitting vibrations generated by the vibratory element **44** to the target treatment area of the subject **48** while also achieving a desirable feature, such as, for example, comfort, noise reduction, intensity, etc. For example, soft and pliable materials such as foam, cloth, or gel may be disposed between the vibratory element **44** and the subject **48** and that transmit vibrations while remaining comfortable to the subject **48**. Increasing thickness of materials of the housing **40** or of materials disposed in the housing **40**, for example, may reduce intensity and/or noise of the vibrations. Alternatively, reducing thickness of the materials of the housing **40** or of materials disposed in the housing **40**, for example, may increase intensity and reduce the profile of the device **10**. In some examples, an intensity of the vibrations may be modified by changing the shape of the housing **40** at the point of contact with the subject's body. In some examples, the housing **40** may be a silicone, plastic, or rubber material. In the illustrated example, the housing **40** is round with smooth surfaces. However, in other examples, the housing **40** may be another shape with textured or bumpy surfaces.

[0128] As shown in FIG. 2, the electrical contacts **18**, **26** and vibration mechanism **28** are placed at various locations on a pelvic region **46** of a subject **48**. The contact pad **42** of the first contact **18** is coupled to a first area **50** of the subject's pelvic region **46**, and specifically adjacent to the inguinal canal of the subject's right side to stimulate an iliohypogastric nerve. The vibration mechanism **28** is coupled to a second area **54** of the subject's pelvic region **46**, specifically on a right testicle of the scrotum **60**, to stimulate a genitofemoral nerve of the subject **48**. And the contact pad **42** of the second contact **26** is coupled to a third area **58** of the subject's pelvic region **46**, and specifically adjacent to an inguinal canal of the subject's lower left side to stimulate both an ilioinguinal nerve and the iliohypogastric nerve.

[0129] The stimulation contacts **18**, **26**, **28** may be coupled to the subject at a combination of locations to best stimulate the testicular cord, including the testicular cremasteric muscles, inguinal canal, ilioinguinal nerve, the iliohypogastric nerve, and/or iliohypogastric nerve of the subject. In FIG. 2, first and second dashed areas **56A**, **56B** represent additional areas for placing one or more contacts **18**, **26**, **28** to treat testicular pain. It will be appreciated that the stimulation contacts **18**, **26**, **28** may be attached to any area of the pelvic region **46**, such as, for example, along a circumference and length of the scrotum **60**.

[0130] In the illustrated example of FIGS. 1 and 2, the first contact **18** has a large rectangular electrode pad **42** and the second contact **26** has a small rectangular electrode pad **42**. The vibration mechanism **28** is puck shaped. The stimulation contacts **18**, **26**, **28** may be selected from a variety



of contacts of various sizes and shapes depending on the treatment area and physiology of the subject. Once selected, the plug **38** of each contact **18, 26, 28** may then be coupled to a socket **70** of the housing **32** before initiating treatment. Each contact pad **42** of the electrical contacts **18, 26** has a contact surface that is configured to adhere to a skin surface of the subject **48**. Similarly, a surface of the vibration mechanism **28** has a contact surface that is configured to adhere to the skin surface of the subject **48**. The contact surfaces may include a textured surface, adhesive, or other material that is configured for self-adhering or otherwise attaching to a skin surface securely (e.g., by suction or friction). Additionally, the contact surface may be removed and reattached to a skin surface repeatedly.

[0131] Each of the wires **34** of the stimulation contacts **18, 26, 28** is removably coupled to a socket **70** formed in the housing **32**. So configured, a subject can customize the stimulation device **10** by attaching a suitable contact of size and shape, and number of contacts. A plurality of sockets **70** are available for coupling additional contacts if more than three stimulation contacts are desired for treatment. In the illustrated example, the plug **38** of the vibration mechanism **28** electrically couples with any of the sockets **70** in the housing **32**. However, in other examples, the vibration mechanism **28** may plug into a special socket **70** suitable only for vibration mechanisms.

[0132] While the stimulation device **10** of FIGS. **1** and **2** includes stimulation contacts **18, 26, 28** that each have a different shape and size, other configurations of the stimulation device **10** are possible. In some examples, the stimulation contacts may be the same size and shape, and may be a different shape than depicted in FIGS. **1** and **2**. For example, to treat a wide area, one or more of the electrical contacts **18, 26** may be long rectangles that extend partially or entirely across a subject's pelvis. In other examples, the electrical contacts **18, 26** may be combined to create a variety of different geometries. In other examples, more than one vibration mechanism **28** may be coupled to the housing **32** to apply vibrations over a larger area or at different areas of the pelvic region **46**.

[0133] Turning back to FIG. **1**, the housing **32** of the device **10** houses the generator **14**, power source **30**, and control system **31**, and defines a plurality of sockets **70** and a plurality of buttons **74**. The control system **31** can communicate through signals (e.g., wired or wirelessly) to one or more components of the device **10**. The controller **31** includes a processor **62** and a memory **66** that stores executable instructions that, when executed by the processor **62**, can control certain settings (e.g., enable features, disable features, active mode, sleep mode, etc.) and parameters (e.g., frequency, intensity, time of delivery, delivery modes, etc.) of the stimulation device **10**. For example, energy parameters may be adjusted to provide a deliverable electrical energy and vibrations having a desired intensity, frequency, amplitude, and duration. In some aspects, the control system **31** is a micro-processor based control system (or controller) that includes, e.g., hardware processor(s), memory module(s), and instructions executable as software code to cause the processor(s) to perform operations to control one or more components of the device **10**. However, the control system **31** can also be realized as a mechanical, electro-mechanical, hydraulic, pneumatic, or other form of a control system or controller without departing from the scope of this disclosure.

[0134] The buttons **74** partially extend outside of the housing **32** for user interaction, and are configured to transmit a variety of different signals to the controller **31** to operate the stimulation device **10**. For example, one or more buttons **74** may be pressed to turn on/off the device **10**, increase or decrease a treatment parameter (e.g., pulse width, intensity, frequency, amplitude, time) of the electrical stimulation contacts **18, 26** and/or the vibration mechanism **28**, or to adjust or program various settings and/or parameters of the stimulation device **10** (e.g., duration of delivering electrical energy, duration of delivering vibrations, setting alarms, providing modes, or programming treatment plans). In some examples, one or more dials, knobs, switches, may be used with or instead of the buttons **74** to operate the device **10**.

[0135] The electrical contacts **18, 26** are configured to deliver electrical energy to the subject **48** via

electrical pulses. The pulses can be adjusted by the subject or caregiver to a desired frequency, pulse width, amplitude, duration (i.e., time of the entire treatment), and mode (e.g., burst, continuous, other pattern) in accordance with a treatment plan and/or a comfort level of the subject **48**.

[0136] For example, the frequency, or the number of electrical pulses delivered in one second, may be selected from a range of approximately 10 Hz or more (e.g., about 15 Hz or more, about 20 Hz or more, about 25 Hz or more, about 30 Hz or more, about 35 Hz or more, about 40 Hz or more, or about 45 Hz) to approximately 100 Hz or less (e.g., about 95 Hz or less, about 90 Hz or less, about 85 Hz or less, about 80 Hz or less, about 75 Hz or less, about 70 Hz or less, about 65 Hz or less, about 60 Hz or less, about 55 Hz or less, about 50 Hz or less, or about 45 Hz). A pulse width, or the elapsed time of a single pulse of energy, may be selected from a range of approximately 50  $\mu$ s or more (e.g., about 60  $\mu$ s or more, about 70  $\mu$ s or more, about 80  $\mu$ s or more, about 90  $\mu$ s or more, about 100  $\mu$ s or more, about 110  $\mu$ s or more, about 120  $\mu$ s or more, about 130  $\mu$ s or more, about 140  $\mu$ s or more, about 150  $\mu$ s or more, about 160  $\mu$ s or more, about 170  $\mu$ s or more, about 180  $\mu$ s or more, about 190  $\mu$ s or more, or about 200  $\mu$ s) to approximately 400  $\mu$ s or less (e.g., about 390  $\mu$ s or less, about 380  $\mu$ s or less, about 370  $\mu$ s or less, about 360  $\mu$ s or less, about 350  $\mu$ s or less, about 340  $\mu$ s or less, about 330  $\mu$ s or less, about 320  $\mu$ s or less, about 310  $\mu$ s or less, about 300  $\mu$ s or less, about 290  $\mu$ s or less, about 280  $\mu$ s or less, about 270  $\mu$ s or less, about 260  $\mu$ s or less, about 250  $\mu$ s or less, about 240  $\mu$ s or less, about 230  $\mu$ s or less, about 220  $\mu$ s or less, about 210  $\mu$ s or less, or about 200  $\mu$ s). An amplitude of the pulse may be selected from a range of approximately 1 mA or more (e.g., about 5 mA or more, about 10 mA or more, about 15 mA or more, about 20 mA or more, about 25 mA or more, about 30 mA or more, about 35 mA or more, about 40 mA or more, about 45 mA or more, or about 50 mA) to approximately 100 mA or less (e.g. about 95 mA or less, about 90 mA or less, about 85 mA or less, about 80 mA or less, about 75 mA or less, about 70 mA or less, about 65 mA or less, about 60 mA or less, about 55 mA or less, or about 50 mA).

[0137] The vibration mechanism **28** can be adjusted by the subject, caregiver, or healthcare provider to a desired frequency or shift in frequency (e.g., gradual change or shift between two or more frequencies), duration (e.g., time of the entire treatment, time of applying vibrations, time of rest, and combination thereof), amplitude, and mode (e.g., burst, continuous, intermittent, or other pattern) in accordance with a treatment plan and/or a comfort level of the subject. The vibrational frequency may be selected from a range of approximately 10 Hz or more (e.g., about 15 Hz or more, about 20 Hz or more, about 25 Hz or more, about 30 Hz or more, about 35 Hz or more, about 40 Hz or more, or about 45 Hz) to approximately 100 Hz or less (e.g., about 95 Hz or less, about 90 Hz or less, about 85 Hz or less, about 80 Hz or less, about 75 Hz or less, about 70 Hz or less, about 65 Hz or less, about 60 Hz or less, about 55 Hz or less, about 50 Hz or less, or about 45 Hz).

[0138] In other examples, the housing **32** may include a user interface (e.g., a display screen, touchscreen, etc.) instead of or in addition to buttons **74** for operating and/or adjusting the parameters and settings of the stimulation device **10**. As will be described in detail below, a subject **48** may adjust the parameters and settings of the stimulation device **10** through a personal electronic device, such as a smartphone, smartwatch, laptop, or other remote computer, that can communicate wirelessly (e.g., via Bluetooth) with the controller **31** disposed in the housing **32**.

[0139] To treat testicular pain in the subject, a healthcare provider may first identify a location of testicular pain. After identifying the location of pain, the healthcare provider can recommend a target treatment area or location on the pelvic region **46** (including the scrotum **60**) to treat the identified location of testicular pain. In some examples, the area or location of pain may be in a different area or location than the target treatment area or region.

[0140] The subject **48** or caregiver can apply the stimulation contacts **18**, **26**, **28** to various areas of the pelvic region **46** by adhering the contacts to the subject's skin. Once in place, the device **10** can be switched on, either wirelessly or locally via the buttons **74**, and a treatment program may begin. Electrical energy and vibrations are delivered to the pelvic region **46** through the stimulation

contacts **18**, **26**, **28** by operating the device **10**. The subject **48** or caregiver can adjust the energy parameters of the deliverable electrical energy to operate and manage an electrical and vibrational stimulation treatment plan using the stimulation device **10**. The stimulation device **10** can operate for a predetermined duration, e.g., 5 minutes to 60 minutes, or can operate in a single delivery of an electrical pulse, shock, or vibration to the target areas.

[0141] Externally coupling the electrical energy source using the device **10** of FIGS. **1** and **2**, for example, includes attaching one or more electrical contacts **18**, **26** to the subject **48**. Coupling the vibration mechanism using the device **10** of FIGS. **1** and **2**, for example, includes attaching or engaging the vibration mechanism **28** to the subject **48**. Once attached, the energy source (e.g., the generator **14**) delivers electrical energy and the vibratory element **44** delivers vibrations to one or more areas of the pelvic region **46**. For example, using the device **10** of FIGS. **1** and **2**, electrical energy is delivered to the subject **48** through multiple electrical pulses through the electrical contacts **18**, **26**, and vibrations are delivered to the subject **48** through the vibration mechanism **28**.

[0142] The electrical energy source may be any suitable method that generates and/or transmits energy to a localized area. In one example, externally coupling an electrical energy source includes engaging an electrode with a skin surface of the subject **48**. In other examples, this may include attaching a first electrode to a first area of the pelvic region **46** and attaching a second electrode to a second area of the pelvic region **46**. In some examples, attaching an electrode may include adhering a first electrode to a region adjacent to an inguinal nerve or canal of the subject **48**. In some examples, the electrode may be coupled to a scrotum **60** of the subject **48**. In yet other examples, the energy source may be something other than an electrode, such as a defibrillator or similar energy delivery device.

[0143] Delivering energy to one or more areas of the pelvic region **46** may include delivering an electrical pulse to the inguinal canal, testicular cord, including the testicular cremasteric muscles, a genitofemoral nerve, an ilioinguinal nerve, and/or an iliohypogastric nerve of the subject **48**. Delivering electrical energy may include delivering multiple electrical pulses over a period of time or delivering a single electrical shock to one or more areas of the pelvic region **46**. In some examples, delivering multiple electrical pulses may include constant stimulation, infrequent bursts, or a combination of modes and methods.

[0144] An example electrical treatment plan for one or more electrical contacts may include delivering a first electrical pulse at a lower frequency for a period of time, and gradually or periodically, increasing the frequency. Another example treatment may include applying intermittent pulses so that the electrical contact delivers pulses at a first frequency for a first time period, and stops for a second time period, and delivers an electrical pulse at the first or different frequency for a third time period after rest. Another example treatment may be fully customizable by the subject. These treatment plans may be done in unison or independently from the other electrical contacts and/or vibration mechanisms.

[0145] An example vibrational treatment plan may include applying a vibration at a lower frequency for a period of time, and gradually or periodically, increasing the frequency. Another example treatment may include applying intermittent vibrations so that the vibration mechanism applies vibrations at a first frequency for a first time period, and stops for a second time period, and applies vibrations at the first or different frequency for a third time period after rest. Another example treatment may be fully customizable by the subject. The stimulation device **10** can operate for a predetermined duration at one or more frequencies, e.g., 5 minutes at 25 Hz, 5 minutes at 50 Hz, etc. In some examples, the device **10** may include a second vibration mechanism so that a first vibration mechanism can be placed on a right side of the pelvic region **46**, and close to the iliohypogastric nerve **50**, and a second vibration mechanism is placed below a scrotum **60** and close to the genitofemoral nerve **54**. A wide variety of combinations of the device **10** are possible.

[0146] These treatment plans may be performed in unison or independently from the other electrical contacts and vibration mechanisms. In some examples, the controller of the stimulation

devices may be configured to selectively adjust the device parameters for each stimulation contact. For example, if a first area of the pelvic region requires more intense treatment than a second treated area, the user can adjust the parameters and/or settings of the electrical contacts and/or vibration mechanisms so that the first electrical contact and vibration mechanism deliver an intense stimulation treatment to a first region and the second electrical contact delivers a less intense stimulation treatment.

[0147] In the example shown in FIGS. 1 and 2, the vibration mechanism 28 is powered by the battery 30 in the device housing 32. However, in other examples, such as the example shown in FIGS. 3A and 3B, a vibration mechanism 128 includes a battery 180 disposed within a housing 140 of the vibration mechanism 128. In some examples, the vibration mechanism 128 can communicate wirelessly with the controller 31 of the device 10 through wireless communication signals 100. In other examples, the vibration mechanism 128 may include a separate controller 186 that can be operated independently from the controller 31 of the device 10.

[0148] Referring to FIGS. 3A and 3B, a second example stimulation assembly 110 includes the stimulation device 10 of FIGS. 1-2 and a wireless vibration mechanism 128. The stimulation device 10 is similar to the stimulation device 10 of FIGS. 1 and 2, and the electrical energy source includes two electrical contacts 18, 26 for delivering electrical energy to a subject 48 and a vibration mechanism 128 for delivering vibrations to the subject 48. Thus, for ease of reference, and to the extent possible, the same or similar components of the assembly 110 will retain the same reference numbers as outlined above with respect to the device 10, although the reference numbers will be increased by 100.

[0149] The vibration mechanism 128 includes the housing 140, a vibratory element 144, the battery 180, and a communication module (e.g., via transmitters and receivers) or controller 186. The vibratory element 144, battery 180, and controller 186 are disposed in the housing 140 and are electrically connected so that the controller 186 can operate the vibratory element 144 and the battery 180 can provide power to the vibratory element 144 and the controller 186. To couple the vibration mechanism 128 to the subject 48, one or more surfaces of the housing 140 may adhere to the skin of the subject. In other examples, the user may hold the vibration mechanism 128 against the treatment area. In some examples, a retractable cord for clipping onto clothing of the subject may be at least partially stored in the housing 140.

[0150] In some examples, the vibration mechanism 128 and the controller 31 of the stimulation device 10 are in wireless communication. In this example, the vibration mechanism 128 is configured to send signals to, and receive signals from, the controller 31 of the device 10, and the controller 31 operates the vibration mechanism 128.

[0151] In some examples, the controller 186 of the vibration mechanism 128 may be independently controlled by one or more buttons 194 or through wireless communication with a personal electronic device, such as a smartphone, smartwatch, laptop, or other remote computer. The control system 186 can communicate through signals (e.g., wired or wirelessly) to one or more components of vibration mechanism 128. The control system 186 may also communicate to the controller 31 of the device 10. The controller 186 includes a processor 196 and a memory 198 that stores executable instructions that, when executed by the processor 196, can control certain settings and parameters of the vibration mechanism 128. For example, treatment parameters may be adjusted to provide vibrations having a desired intensity, frequency, and duration. In some aspects, control system 186 is a micro-processor based control system (or controller) that includes, e.g., hardware processor(s), memory module(s), and instructions executable as software code to cause the processor(s) to perform operations to control one or more components of the vibration mechanism 128. A plurality of buttons 194 are configured to receive input to operate and control the vibration mechanism 128. In some examples, the controller 186 may be operated (e.g., switch the device on or off, adjust vibration intensity, adjust vibration timing, etc.) by pressing one or more of the buttons 194. However, the control system 186 can also be realized as a mechanical, electro-

mechanical, hydraulic, pneumatic, or other form of a control system or controller without departing from the scope of this disclosure. One or more vibrational treatment programs may be stored in the memory of the processor can be accessed and adjusted.

[0152] The buttons **194** partially extend outside of the housing **140** for user interaction, and are configured to transmit a variety of different signals to the controller **186** to operate the vibration mechanism **128**. For example, one or more buttons **194** may be pressed to turn on/off the vibration mechanism **128**, increase or decrease a treatment parameter (e.g., intensity, frequency, time, etc.), or to adjust or program various settings of the vibration mechanism **128** (e.g., duration of delivering vibrations, setting alarms, providing modes, or programming treatment plans). In some examples, one or more dials, knobs, switches, may be used with or instead of the buttons **194** to operate the vibration mechanism **128**.

[0153] In some examples, the housing **140** of the vibration mechanism **128** may include a user interface (e.g., a display screen, touchscreen, etc.) instead of or in addition to buttons **194** for operating and/or adjusting the treatment parameters and settings of the vibration mechanism **128**. For example, a subject, caregiver, or healthcare provider may adjust the treatment parameters and settings of the vibration mechanism **128** through a personal electronic device, such as a smartphone, smartwatch, laptop, or other remote computer that can communicate wirelessly (e.g., via Bluetooth) with the controller **186** disposed in the housing **140**. In some examples, the healthcare provider may monitor the subject's usage of the device and adjust the treatment parameters from a remote location.

[0154] While the vibration mechanism **128** shown in FIG. 3B has been illustrated as having the vibratory element **144** centrally disposed in the housing **140** between parallel sides **116**, **117**, in other examples, the vibratory element may be disposed closer to one side of the housing **140**. In this example, engaging this one side of the housing with the subject may apply stronger vibrations than engaging the other side of the housing with the subject. In other examples, the vibration mechanism **128** may include a first vibratory element adjacent to one side of the housing and a second vibratory element adjacent to the other side of the housing. In this case, either side or surface of the housing can be coupled to (or engaged with) the subject to apply vibrations interchangeably.

[0155] Referring now to FIGS. 4-7, an electrical stimulation device **210** includes an electrical energy source and vibration source for delivering electrical and vibrational energy to a pelvic region **46** of a subject **48**. The stimulation device **210** of FIGS. 4-7 is similar to the stimulation device **10** of FIGS. 1 and 2, and includes two electrical contacts **218**, **226** for delivering electrical energy and a vibration mechanism **228** for delivering vibrations to the subject **48**. Thus, for ease of reference, and to the extent possible, the same or similar components of the device **210** of FIGS. 4-7 will retain the same reference numbers as outlined above with respect to the device **10** of FIGS. 1 and 2, although the reference numbers will be increased by 200.

[0156] The stimulation device **210** includes a flexible body **212** including an interior surface **216** (FIG. 4), an exterior surface **217** (FIG. 5), two electrodes **218**, **226**, and a vibration mechanism **228** attached to (e.g., embedded between) one or more layers of the body **212**. A housing **232** contains an electrical signal generator **214**, a battery **230**, and a control system **231** (FIG. 7). In FIG. 4, each contact pad **242** of the electrodes **218**, **226** is exposed on the interior surface **216** of the body **212**, and in FIG. 5, the electrodes **218**, **226** are covered by a layer of material on the exterior surface **217**. The vibration mechanism **228** has an exterior surface that can engage the subject directly **48** or may be covered by one or more layers of material. As shown in FIG. 6, the body **212** can be arranged into a cuff that is sized to fit at least partially around a scrotum **60** of the subject **48**. The cuff is arranged so that the interior surface **216** engages the scrotum **60** of the subject **48**, as shown in FIG. 7. In this way, the cuff can stimulate the testicular cord and cremasteric muscle of the subject **48** to treat testicular pain.

[0157] The electrodes **218**, **226** and vibration mechanism **228** are spaced relative to each other in a

way that when the cuff is positioned on the scrotum **60**, the contact pads **242** of the electrodes **218**, **226** and vibration mechanism **228** engage different areas of the subject's scrotum **60**. When arranged on the scrotum **60**, the first and second electrodes **218**, **226** are placed on a back side of the scrotum **60**, and the vibration mechanism **228** is placed on a front side of the scrotum **60**. However, the cuff may be rotated to stimulate different areas of the scrotum **60**. In other examples, the body **212** may include additional electrodes or vibration mechanisms placed variously along a length of the body **212**. The electrodes **218**, **226** and vibration mechanism **228** are electrically coupled by a wire **278** embedded between one or more layers of the body **212**. The wire **278** extends externally from the body **212** to connect with a housing **232** (FIG. 7). The device **210** may be operated in the same or similar manner as the device **10** by adjusting the settings of the device **210** and/or treatment parameters using the buttons **274** and/or the control system **231**. In some examples, the vibration mechanism **228** may be battery powered.

[0158] The device **210** includes an adjustable coupler to treat a range of scrotal sizes. The coupler includes a first coupling **282** disposed at a first end **286** of the body **212** and a second coupling **290** disposed at a second end **294** of the body **212**. The first and second couplings **282**, **290** are disposed on opposite surfaces **216**, **217** of the body **212** such that when the first and second ends **286**, **294** of the body **212** are joined to form the cuff, the first and second couplings **282**, **290** overlap and mate. In the illustrated example, the coupler is a hook-and-loop fastener where the first coupling **282** includes a plurality of hooks that are arranged to fasten to a plurality of loops of the second coupling **290**.

[0159] In some examples, the body **212** of the stimulation device **210** may have a microcontroller, power source, and/or generator integrated into one or more layers of the body **212**. In such an example, the body **212** may be rechargeable and may be configured to directly communicate with a user's personal electronic device for operation.

[0160] In some examples, the device **210** may include more or fewer electrodes and more vibration mechanisms that can be activated or deactivated depending on the treatment plan to treat testicular pain.

[0161] In some examples, the body **212** of the stimulation device **210** may be arranged differently so that the cuff is formed using, for example, a tie, buckle, button, snap, or other suitable male-female mating structure or fastener. In other examples, the body **212** is circular and only needs slight adjustment after placing the device **210** around the scrotum **60** of the subject. In this case, rather than coupling opposing ends of a rectangular strip embedded with electrodes and a vibration mechanism, one example device may be preconfigured as a cuff to receive the scrotum into a central opening of the cuff. In some examples, the body **212** may be made of a memory-shape material that can hold a new shape in response to being manipulated and shaped (e.g., bent, twisted, etc.).

[0162] In some examples, the stimulation device **10**, **210** may be incorporated into a garment, such as underwear, as illustrated in FIG. 8. For example, an underwear pouch that covers a subject's penis and scrotum may include a plurality of spaced electrodes and vibration mechanisms that can be selectively operated to deliver electrical and vibrational stimulation to various locations of the pelvic region. Other regions of the underwear may also incorporate electrical energy and vibrational sources.

[0163] Referring now to FIG. 8, a treatment system **340** includes a wearable assembly **310**, a smartphone **350**, and a stimulation device (hidden from view), such as the devices **10**, **210** or assembly **110** described herein. The stimulation device is removably coupled to a garment **300**, which in this example is underwear. The undergarment **300** has a built-in pocket **308** sized to receive the housing (e.g., housing **232** of a device **210**) and accommodates a cuff, contacts, and wires. The garment **300** fits over the stimulation device **10**, **210** attached to the subject **48** so that the subject **48** may wear and easily operate the stimulation device **10**, **210** throughout the course of a day.

[0164] The devices **10, 210** described herein may be operated using a personal electronic device **350**, such as a smartphone, smartwatch, laptop, or other remote computer, as shown in the example treatment system **340** of FIG. **8**. The treatment system **340** includes an application stored on a personal electronic device **350**. Using wireless communication, such as via Wi-Fi, the subject **48** can control the stimulation device **210** remotely via wireless communication signals **399** rather than retrieving the housing **232** from the pocket **308** of the garment **300** and operating the device manually (although, in some examples, manual operation is optional). In this example, the subject **48** can use an application stored on the personal electronic device **350** to operate and adjust various features of the stimulation device. For example, the subject **46** can adjust the settings and parameters of the stimulation device to deliver a particular stimulating treatment to the pelvic region **46**. The application may also be programmable so that the subject **48** may create and store a treatment plan for repeated use. The application may also be remotely accessed by a health care provider so that the health care provider may adjust the settings and/or parameters of the device **10, 210** and create and/or modify treatment programs customized for the subject **48**.

[0165] In the illustrated example of FIG. **8**, the personal electronic device **350** displays two energy parameters for controlling an energy output of a first electrode. However, in other examples, each electrode and vibration mechanism may have more or fewer adjustable parameters and settings. Additionally, in some examples, the energy parameters of the electrodes and/or vibration mechanisms may be adjusted collectively.

[0166] While the stimulation devices and assemblies **10, 110, 210** of FIGS. **1-7** have been described as having a vibration mechanism **28, 128, 228** that adheres or engages directly to a subject's skin, in some examples, a vibration mechanism may hang from a belt or strap, as shown in FIGS. **9-11**.

[0167] Referring to FIG. **9**, a stimulation assembly **410** includes a vibration mechanism **428** and the device **10, 210** for delivering vibrational and electrical energy to a pelvic region **46** of a subject **48** to treat testicular pain. The stimulation assembly **410** is similar to the stimulation assembly **110** of FIGS. **3A** and **3B**, and includes one or more electrical contacts for delivering electrical energy and the vibration mechanism **428** for delivering vibrations to the subject **48**. Thus, for ease of reference, and to the extent possible, the same or similar components of the assembly **410** of FIG. **9** will retain the same reference numbers as outlined above with respect to the assembly **110** of FIGS. **3A** and **3B**, although the reference numbers will be increased by 300.

[0168] In FIG. **9**, the vibration mechanism **428** is coupled to the subject **48** by an adjustable strap or belt **418** extending around a waist or hips of the subject **48**. The vibration mechanism **428** is positioned close to an iliohypogastric nerve on a right side of the subject's pelvic region **46**. The user may lower the belt **418** (or strap the belt **418** around the subject's thigh **49**) to position the vibration mechanism **428** close to an ilioinguinal nerve or close to a genitofemoral nerve.

[0169] The vibration mechanism **428** is puck-shaped and can be positioned at any location on the belt **418** by sliding the vibration mechanism **428** relative to the belt **418** and/or by clipping the vibration mechanism **428** to the belt **418** at a desired location. The vibration mechanism **428** is fastened (e.g., clipped, snapped, buckled, hook-and-loop mated, etc.) to the belt **418** so that the vibration mechanism **428** is positioned in a desired target area of treatment. In some examples, the user may rotate the belt **418** relative to their body and vibration mechanism **428** to position the vibration mechanism **428** at a target area. In some examples, the vibration mechanism **428** is rotatable relative to the belt **418** to position the vibration mechanism **428**. In some examples, the vibration mechanism **428** may be adjusted vertically (relative to the orientation of FIG. **9**) so that the belt **418** remains around the user's waist or hips while the vibration mechanism **428** is positioned at a lower portion of the subject's pelvic region **46**. While not illustrated in FIG. **9**, the vertical positioning of the vibration mechanism **428** may be adjusted using a retractable cord or additional strap that is coupled to the vibration mechanism **428**. In another example, the belt **418** may be fastened to an upper thigh **49** of the subject **48**. The belt **418** is adjustable to accommodate

different body sizes and different body parts. The belt **418** may be a pliable or stretchy material. [0170] In FIG. **10**, another example stimulation assembly **510** includes a vibration mechanism **528** and the device **10**, **210** for applying vibrational and electrical energy to a pelvic region **46** of a subject **48** to treat testicular pain. The stimulation assembly **510** is similar to the stimulation assembly **110** of FIGS. **3A** and **3B**, and includes one or more electrical contacts for delivering electrical energy and a vibration mechanism **528** for delivering vibrations to the subject **48**. Thus, for ease of reference, and to the extent possible, the same or similar components of the assembly **510** of FIG. **10** will retain the same reference numbers as outlined above with respect to the assembly **110** of FIGS. **3A** and **3B**, although the reference numbers will be increased by 400.

[0171] The assembly **510** is configured to be worn under a garment **500**, such as shorts or briefs, and the vibrational mechanism **528** is attached to a waistband of the garment **500** or a belt **518**. The vibration mechanism **528** of the assembly **510** is coupled to a strap **520**, and is interiorly placed relative to the garment **500** worn by the subject **48**. The strap **520** has an adjustment mechanism **524** that allows the strap **520** to extend or contract based to a desired length. At a distal end **522**, the strap **520** forms a hook so that an underside surface of the hook engages an exterior surface of the belt **518** or waistband of the garment **500** and secures the vibration mechanism **528** to the subject's clothing. The underside of the hook **522** has a grippable surface that securely attaches the strap **520** to the garment **500** and/or belt **518**, but is configured for easily disengaging from the garment **500** and/or belt **518** to reposition or remove the vibration mechanism **528**. The vibration mechanism **528** may be placed directly on the skin of the subject **48**, or may be placed between a layer of material and the garment **500**.

[0172] Referring to FIG. **11**, another example stimulation assembly **610** includes a vibration mechanism **628** and the device **10**, **210** for applying vibrational and electrical energy to a pelvic region **46** of a subject **48** to treat testicular pain. The stimulation assembly **610** is similar to the stimulation assembly **110** of FIGS. **3A** and **3B**, and includes one or more electrical contacts for delivering electrical energy and a vibration mechanism **628** for delivering vibrations to the subject **48**. Thus, for ease of reference, and to the extent possible, the same or similar components of the assembly **610** of FIG. **11** will retain the same reference numbers as outlined above with respect to the assembly **110** of FIGS. **3A** and **3B**, although the reference numbers will be increased by 500.

[0173] The stimulation assembly **610** is coupled to a different garment **600**, such as underwear, and is configured to apply vibrations over a layer of material (e.g., the underwear material). The garment **600** includes a reinforced opening **640** that is sized to receive a distal end of a strap **620** to hook onto the undergarment **600**. In this example, the garment **600** is configured for receiving the vibrational mechanism **628**, and treating testicular pain, on a right side of the subject **48**. In other examples, the garment **600** may be configured for treating testicular pain on a left side of the subject **48** by providing a reinforced opening on the left side of the garment **600**. In some examples, the garment **600** may include two or more openings for repositioning the vibration mechanism **628** as needed.

[0174] While the stimulation devices and assemblies **10**, **110**, **310**, **410**, **510**, **610** of FIGS. **1-11** have been depicted as having a single vibration mechanism **28**, **128**, **228**, **428**, **528**, **628**, in some examples, a stimulation device or assembly may include more than one vibration mechanism, as shown in FIG. **12**.

[0175] In FIG. **12**, an example stimulation assembly **710** includes two vibration mechanisms **726**, **728** and the device **10**, **210** for applying vibrational and electrical energy to a pelvic region **48** of a subject **46** to treat testicular pain. The stimulation assembly **710** is similar to the stimulation assembly **110** of FIGS. **3A** and **3B**, and includes one or more electrical contacts for delivering electrical energy and a vibration mechanism for delivering vibrations to the subject **48**. Thus, for ease of reference, and to the extent possible, the same or similar components of the assembly **710** of FIG. **12** will retain the same reference numbers as outlined above with respect to the assembly **110** of FIGS. **3A** and **3B**, although the reference numbers will be increased by 600.



[0176] In FIG. 12, the wearable assembly 710 includes a garment 700, which in this example is underwear, and two vibration mechanisms 726, 728 coupled to the garment. The undergarment 700 has two, internally built-in pockets 706, 708, with each pocket 706, 708 being sized to receive one of the vibration mechanisms 726, 728. Each of the vibration mechanisms 726, 728 includes a vibratory element 744, a battery, and a controller or communication module 786, 787. Accordingly, the vibration mechanisms 726, 728 can be programmed to apply different vibrational massage treatments to the different areas of the pelvis 46. The vibration mechanisms 726, 728 may also be programmed to apply the same vibrational massage treatment.

[0177] Each vibration mechanism 726, 728 of the stimulation assembly 710 may be activated or deactivated depending on the treatment plan to treat testicular pain. In some examples, each of the controllers 786, 787 of the vibration mechanisms 728, 726 may be configured to selectively adjust the treatment parameters for the respective vibratory element 744. For example, if a first area of the pelvic region 46 requires more intense treatment than a second treated area, the user can adjust the settings and/or parameters of the vibration mechanism 726 so that the first vibration mechanism 726 delivers an intense massage vibration treatment to the first region, and adjust the settings and/or parameters of the vibration mechanism 728 so that the second vibration mechanism 728 delivers a less intense massage vibration treatment.

[0178] The subject 48 may wear and easily operate the stimulation device 310 throughout the course of a day. For example, the user can retrieve one or more of the vibration mechanisms 726, 728, operate it (e.g., turn on/off, adjust vibration parameters via the buttons), and insert the one or more vibration mechanism 726, 728 back into the pockets 706, 708. In other examples, the undergarment 700 may include more or fewer pockets. For example, the underwear pouch that covers a subject's penis and scrotum may include a plurality of pockets for receiving a plurality of vibration mechanisms that can be selectively operated to deliver vibrations to various locations of the pelvic region 46. In another example, the pockets 706, 708 may be large enough to receive two or more vibration mechanisms.

[0179] While the stimulation device 710 of FIG. 12 has been described as having two vibration mechanisms 726, 728, in other examples, a wearable stimulation device or assembly may include one or more than two vibration mechanisms that fit into a corresponding number of pockets disposed in the same or different areas relative to the pelvic region of the garment. In other examples, the vibration mechanisms may be sewn into the undergarment for permanent fixation.

[0180] While the stimulation device 710 of FIG. 12 has been described as having two separate and independently controlled vibration mechanisms 726, 728, in other examples, a stimulation assembly or device may include two or more vibration mechanisms that are connected and do not operate independently. For example, one or more vibration mechanisms may be electrically coupled to the device 10, 210. In another example, the vibration mechanisms may be electrically connected by a wire that can stretch or extend between the pockets of the garment.

[0181] While the vibration mechanisms 28, 128, 228, 428, 528, 628, 726, 728 of the stimulation devices and assemblies 10, 110, 210, 310, 410, 510, 610, 710 of FIGS. 1-12 have been depicted or described as smooth-surfaced pucks, in other examples, a vibration mechanism may have a different shape and the housing may have different surface treatment.

[0182] While the vibration mechanisms 28, 128, 228, 428, 528, 628, 726, 728 of the stimulation devices and assemblies 10, 110, 210, 310, 410, 510, 610, 710 of FIGS. 1-12 have been described as having one vibratory element, in other examples, a vibration mechanism of a stimulation device may have two or more vibratory elements disposed in a vibration mechanism housing or in the device housing.

[0183] In some examples, multiple vibratory elements share electrical power with one another, and in other examples, the vibratory elements may connect to separate batteries. The controller of the device or of the vibration mechanism may be coupled to each vibratory element separately, or in other examples, the controller may be coupled to only one vibratory element, which in turn is

coupled to the other vibratory elements, so that the controller operates the vibratory elements in unison.

[0184] While the stimulation devices and assemblies **310, 410, 510, 610, 710** of FIGS. **8-12** are coupled to, or embedded in, garments (e.g., pants, underwear, briefs, belts), in other examples, a stimulation device may be incorporated in different wearable items. In other examples, the stimulation device may be a handheld device.

[0185] In some examples, the stimulation devices and assemblies **10, 110, 210, 310, 410, 510, 610, 710** may incorporate heating or cooling elements that can be operated to deliver heat or a cooling sensation in addition to or instead of applying vibrations or delivering electrical energy. In some examples, the vibratory element of the vibration mechanism is configured to emanate heat or a cooling sensation while applying vibrations.

[0186] In some examples, treating testicular pain includes delivering electrical energy to a subject **48** and the devices and assemblies **10, 110, 210, 310, 410, 510, 610, 710** may include one or more of the following methodologies for delivery electrical energy: transcutaneous electrical nerve stimulation, Russian stimulation, neuromuscular electrical stimulation (NMES), inferential current (IFC) electrical stimulation, high voltage electrical stimulation, iontophoresis, or a combination thereof.

[0187] In some examples, the stimulation devices and assemblies **10, 110, 210, 310, 410, 510, 610, 710** may be configured for delivering electrical, radiation, magnetic, conducted energy, or a combination thereof to effectively treat testicular pain.

[0188] With reference to FIGS. **1-12**, an example operation of the stimulation devices and assemblies **10, 110, 210, 310, 410, 510, 610, 710** can be performed to treat testicular pain. One or more steps of the example operation can be performed by or with the control system **31, 186, 231, 486, 586, 686, 786, 787** (e.g., with input from a human operator). For instance, the operation can include adjusting a pulse intensity, setting a timer for treatment, sending results to a healthcare provider, etc.

[0189] FIG. **13** shows a schematic drawing of a control system **1000** that can be used, e.g., as control system **31, 186, 231, 486, 586, 686, 786, 787**, to perform operations according to the present disclosure. Some or all of the example control system **1000** can be implemented as cloud-based system and/or service, alone or in combination with other portions of the example control system **1000** that can be implemented to execute. The controller **1000** is intended to include various forms of digital computers, such as printed circuit boards (PCB), processors, digital circuitry, or otherwise. Additionally, the system can include portable storage media, such as, Universal Serial Bus (USB) flash drives. For example, the USB flash drives may store operating systems and other applications. The USB flash drives can include input/output components, such as a wireless transmitter or USB connector that may be inserted into a USB port of another computing device.

[0190] The controller **1000** includes a processor **1010**, a memory **1020**, a storage device **1030**, and an input/output device **1040**. Each of the components **1010, 1020, 1030, and 1040** are interconnected using a system bus **1050**. The processor **1010** is capable of processing instructions for execution within the controller **1000**. The processor may be designed using any of a number of architectures. For example, the processor **1010** may be a CISC (Complex Instruction Set Computers) processor, a RISC (Reduced Instruction Set Computer) processor, or a MISC (Minimal Instruction Set Computer) processor.

[0191] In one implementation, the processor **1010** is a single-threaded processor. In another implementation, the processor **1010** is a multi-threaded processor. The processor **1010** is capable of processing instructions stored in the memory **1020** or on the storage device **1030** to display graphical information for a user interface on the input/output device **1040**.

[0192] The memory **1020** stores information within the control system **1000**. In one implementation, the memory **1020** is a computer-readable medium. In one implementation, the memory **1020** is a volatile memory unit. In another implementation, the memory **1020** is a non-

volatile memory unit.

[0193] The storage device **1030** is capable of providing mass storage for the controller **1000**. In one implementation, the storage device **1030** is a computer-readable medium. In various different implementations, the storage device **1030** may be a floppy disk device, a hard disk device, an optical disk device, a tape device, flash memory, a solid state device (SSD), or a combination thereof.

[0194] The input/output device **1040** provides input/output operations for the controller **1000**. In one implementation, the input/output device **1040** includes a keyboard and/or pointing device. In another implementation, the input/output device **1040** includes a display unit for displaying graphical user interfaces.

[0195] The features described can be implemented in digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. The apparatus can be implemented in a computer program product tangibly embodied in an information carrier, for example, in a machine-readable storage device for execution by a programmable processor; and method steps can be performed by a programmable processor executing a program of instructions to perform functions of the described implementations by operating on input data and generating output. The described features can be implemented advantageously in one or more computer programs that are executable on a programmable system including at least one programmable processor coupled to receive data and instructions from, and to transmit data and instructions to, a data storage system, at least one input device, and at least one output device. A computer program is a set of instructions that can be used, directly or indirectly, in a computer to perform a certain activity or bring about a certain result. A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment.

[0196] Suitable processors for the execution of a program of instructions include, by way of example, both general and special purpose microprocessors, and the sole processor or one of multiple processors of any kind of computer. Generally, a processor will receive instructions and data from a read-only memory or a random-access memory or both. The essential elements of a computer are a processor for executing instructions and one or more memories for storing instructions and data. Generally, a computer will also include, or be operatively coupled to communicate with, one or more mass storage devices for storing data files; such devices include magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and optical disks. Storage devices suitable for tangibly embodying computer program instructions and data include all forms of non-volatile memory, including by way of example semiconductor memory devices, such as EPROM, EEPROM, solid state drives (SSDs), and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, ASICs (application-specific integrated circuits).

[0197] To provide for interaction with a user, the features can be implemented on a computer having a display device such as a CRT (cathode ray tube) or LCD (liquid crystal display) or LED (light-emitting diode) monitor for displaying information to the user and a keyboard and a pointing device such as a mouse or a trackball by which the user can provide input to the computer. Additionally, such activities can be implemented via touchscreen flat panel displays and other appropriate mechanisms.

[0198] The features can be implemented in a control system that includes a back-end component, such as a data server, or that includes a middleware component, such as an application server or an Internet server, or that includes a front-end component, such as a client computer having a graphical user interface or an Internet browser, or any combination of them. The components of the system can be connected by any form or medium of digital data communication such as a

communication network. Examples of communication networks include a local area network (“LAN”), a wide area network (“WAN”), peer-to-peer networks (having ad-hoc or static members), grid computing infrastructures, and the Internet.

[0199] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of any disclosure or of what may be claimed, but rather as descriptions of features that may be specific to particular examples of particular disclosures. Certain features that are described in this specification in the context of separate examples can also be implemented in combination in a single example. Conversely, various features that are described in the context of a single example can also be implemented in multiple examples separately or in any suitable subcombination. Moreover, although features may be described herein as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

[0200] Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system modules and components in the examples described herein should not be understood as requiring such separation in all examples, and it should be understood that the described program components and systems can generally be integrated together in a single product or packaged into multiple products.

[0201] Particular examples of the subject matter have been described. Other examples are within the scope of the following claims. For example, the actions recited in the claims can be performed in a different order and still achieve desirable results. As one example, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain implementations, multitasking and parallel processing may be advantageous.

## Claims

1. A method of treating testicular pain of a subject, the method comprising: coupling an electrical energy source to a pelvic region of the subject; coupling a vibration mechanism to the pelvic region of the subject; delivering electrical energy to the pelvic region; and applying vibrations to the pelvic region.
2. The method of claim 1, wherein coupling the electrical energy source comprises externally engaging the electrical energy source with a scrotum of the subject.
3. The method of claim 1, wherein coupling the vibration mechanism comprises engaging the vibration mechanism with a scrotum of the subject.
4. The method of claim 1, wherein coupling the electrical energy source and coupling the vibration mechanism comprise connecting a stimulation device to the pelvic region of the subject, wherein the stimulation device comprises an electrode and the vibration mechanism.
5. The method of claim 4, wherein coupling the stimulation device comprises attaching a cuff at least partially around a scrotum of the subject, wherein the cuff comprises the electrode and the vibration mechanism, wherein the electrode and the vibration mechanism are arranged to engage the scrotum when the cuff is attached to the scrotum.
6. The method of claim 4, comprising positioning the vibration mechanism in a first area of the pelvic region and positioning the electrode in a second area of the pelvic region.
7. The method of claim 1, wherein coupling the electrical energy source comprises adhering an electrode to the subject.
8. The method of claim 7, wherein delivering the electrical energy comprises delivering an

electrical pulse at a frequency in a range of approximately 0 Hz to approximately 150 Hz.

**9.** The method of claim 7, wherein applying vibrations comprises applying vibrations having a frequency in a range of approximately 0 Hz to approximately 50 Hz.

**10.** The method of claim 1, wherein applying vibrations and delivering the electrical energy comprise delivering an electrical pulse and applying vibrations to a testicular cord, a cremasteric muscle, an inguinal canal, a genitofemoral nerve, an ilioinguinal nerve, and/or an iliohypogastric nerve of the subject.

**11.** A method of treating testicular pain in a subject, the method comprising: coupling a stimulation device to a pelvic region of the subject, the stimulation device comprising an electrical contact configured for delivering electrical energy and a vibration mechanism configured for delivering vibrations; delivering, via the stimulation device, electrical energy to the subject; and delivering, via the stimulation device, vibrations to the subject.

**12.** The method of claim 11, wherein coupling the stimulation device comprises engaging one or more of the electrical contacts and the vibration mechanism to a scrotum of the subject.

**13.** The method of claim 12, wherein coupling the stimulation device comprises positioning a second electrical contact of the stimulation device in an area of the pelvic region that is different than the scrotum of the subject.

**14.** The method of claim 11, wherein coupling the stimulation device comprises fastening a cuff of the stimulation device around a scrotum of the subject, wherein the electrical contact and the vibration mechanism are integrated with the cuff of the stimulation device.

**15.** The method of claim 11, wherein delivering electrical energy comprises delivering an electrical pulse to the subject.

**16.** The method of claim 15, comprising adjusting, via a controller, an intensity of a deliverable electrical energy and/or deliverable vibrations, wherein the controller is operatively coupled to the electrical contact and to the vibration mechanism.

**17.** The method of claim 16, wherein adjusting comprises adjusting a stimulation parameter using a personal electronic device, wherein the controller is communicatively coupled to the personal electronic device.

**18.** A wearable device for treating testicular pain, the device comprising: a first stimulation contact configured for delivering an electrical pulse to a pelvic region of a subject; and a second stimulation contact configured for delivering vibrations to the pelvic region of the subject.

**19.** The device of claim 18, comprising a flexible band arranged for removably attaching the first stimulation contact and the second stimulation contact to a scrotum of the subject, wherein the first stimulation contact and second stimulation contact are coupled to the flexible band.

**20.** The device of claim 18, comprising a controller operatively coupled to the first stimulation contact and the second stimulation contact, the controller comprising a processor, a memory communicatively coupled to the processor and storing executable instructions that, when executed by the processor, causes the processor to: receive data transmitted by a remote communication device; send a signal to a generator coupled to the first stimulation contact; adjust an energy parameter of the electrical pulse generated by the generator; send a signal to a vibratory element of the second stimulation contact; and adjust an intensity parameter of vibrations generated by the vibratory element.

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